Tanish Singh Rajpal

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Research interests: learning and certification methods for reliable quantum computation and quantum chemistry

EDUCATION

Carnegie Mellon University — *Mountain View, CA*
M.S. in Information Technology - Information Security, Information Networking Institute — *Expected May 2027*

NMIMS University — *Shirpur, India*
B.Tech. in Computer Engineering | GPA: 3.83/4.00 | Valedictorian, graduating class — *May 2025*

Relevant coursework: Linear Algebra and Differential Equations, Complex Variables and Transforms, Probability and Statistics, Discrete Mathematics, Quantum and Statistical Physics, Design and Analysis of Algorithms, Theoretical Computer Science, Operations Research, Machine Learning (all A or A+).

RESEARCH EXPERIENCE

Entanglement Delivery Guarantees in Quantum Networks — *May 2026 - present*
Independent Study, CMU Information Networking Institute | Advisor: Prof. Michael Mattarock

- Built a **model-based delivery guard** and a **random-test certificate** for kept-pair average fidelity, with tests that measure how optimistic the model is.
- Ran pre-registered simulations: on 3D and diagonal topologies, **0 bad pairs in 8.8 million deliveries** when the guard declared a safety margin against a model deliberately wrong by 0.044; on a 5×5 grid with an exact model, **0 of 91,392 users** received a bad pair. **83 of 95 primary hypotheses passed**, with all 12 failures reported.
- Re-analyzed **seven experimental datasets**, including Harvard and IonQ: **predicted the IonQ certificate in advance (0.817 vs. 0.818 measured**, 3,000 pairs); Oxford record certified average fidelity **above 0.9 after ~220,000 kept pairs** (final 0.917, trusted settings). Manuscript complete; code and evidence.

Differentiable Quantum-Chemical Models (DFTBML) — *Sep 2026 - present*
Independent Study, CMU Chemistry | Advisor: Prof. David Yaron

- Extending **DFTBML** (Hu, He, and Yaron, *J. Chem. Theory Comput.* 2023), a machine-learned tight-binding model trained on CCSD(T) energies, to a **split-valence (double-zeta) basis** to improve accuracy and transferability.
- Designed a **four-shell basis** in PyTorch, scaling learned parameters from **75 to 266** under symmetry constraints.
- Built a **randomized-perturbation test suite** that catches improperly tied parameters, validated with injected faults.
- Calibrated **PySCF** integrals against reference **Slater-Koster tables** to initialize the new basis functions.

Post-Quantum EAP-TLS for Enterprise Wi-Fi — *Aug 2026 - present*
Master's Thesis, CMU INI | Advisors: Prof. Pedro Bustamante and Prof. Michael Mattarock | Reader: Prof. Patrick Tague

- Built **LockStep**, an ns-3 simulator of the full enterprise Wi-Fi sign-in path (802.1X/EAP-TLS over 802.11ax, RADIUS, IPv6 path MTU, site-to-site VPNs, TCP and QUIC backbones).
- Showed that **ML-DSA certificates raise sign-in round trips from 6 to 29**, and that oversized fragments **fail completely** when firewalls filter ICMPv6 Packet Too Big.
- An **edge QUIC proxy** at the wireless controller cut sign-in time **about 10× (1.50 s to 135 ms** at 50 ms RTT) and stayed robust behind VPNs; now adding a QKD-keyed backbone and 500-device sign-in storms.

TEACHING AND SERVICE

Incoming Head Teaching Assistant, Quantum Security — *Spring 2027*
CMU Information Networking Institute | Instructor: Prof. Michael Mattarock

- Helping Prof. Mattarock build the course ahead of its Spring 2027 offering; will serve as **head TA**.

Student Ambassador, CMU Information Networking Institute — *Oct 2025 - present*

SKILLS

Quantum and Statistics: network simulation, anytime-valid inference, certification, VQE, QAOA, error mitigation.
Software and ML: Python, Qiskit, Cirq, PennyLane, NetSquid, ns-3, Git; reinforcement learning (PPO), deep learning.
Security and Systems: ML-DSA, QKD, 802.1X, penetration testing; C, x86-64 assembly, Linux administration.

PUBLICATIONS

Manuscript

Rajpal, T. S., and Mattarock, M. (2026). "What can a quantum network promise about the entangled pairs it delivers?" Manuscript. Code and evidence.

Journal article

Rajpal, T. S., and Naithani, A. (2026). "NeuroCrypt: A Neuro Symbolic AI Ecosystem for Advanced Cryptographic Data Security and Transmission." *IEEE Transactions on Artificial Intelligence*, 7(1), 512-521. doi:10.1109/TAI.2025.3577605

Conference papers

Naithani, A., and **Rajpal, T. S.** (2025). "Blockchain-Based Digital Health Passport: A Decentralized Framework for Secure and Interoperable Health Data Management." *IEEE ZINC 2025*. doi:10.1109/ZINC65316.2025.11100281

Naithani, A., **Rajpal, T. S.**, and Mistry, V. (2024). "TAV-Cart: Transaction Assistance and Validation Cart System for Effortless Shopping and Checkout." *IEEE ICACCM 2024*. doi:10.1109/ICACCM61117.2024.11059075

Naithani, A., Jain, V., **Rajpal, T. S.**, and Mistry, V. (2024). "AI Based Goalkeeper for Penalty Shot Prediction Using RNN and Pattern Matching Algorithm." *IEEE ISDFS 2024*. doi:10.1109/ISDFS60797.2024.10527307

QUANTUM PROJECTS AND COMPETITIONS

The Last Call - Quantum Interference Horror Game *Oct 2026*

Qollab × IonQ Global Quantum Hackathon | Global Best of Quantum Playground | CMU node first place | Team Popeyes

- Built an AR horror game on the interference vs. which-path trade-off: trace interference to the right door, lose it, then erase a record to restore it. Website.
- Also won Most Novel/Creative and Best Documentation at the CMU node (12 projects).

Dicke-State Circuits for Google Willow *Oct 2026*

DuQuantum 2026 (Duke), Google Quantum AI Challenge | Third place, Standard Track | Team Popeyes

- Prepared **Dicke states** on Google's **105-qubit Willow** layout using only CZ and PhasedXZ gates; the code picks a recipe by size (W-state tree, exact small templates, connected-shape portfolio, Bärtschi-Eidenbenz fallback).
- Placed circuits using the day's **calibration data**, avoiding unusable qubits and couplers. Website.

Phase Hunter - Mapping Quantum Spin Phases under Noise *Sep 2026*

Team Lead, Q-SITE Hacks 2026 | Quantum Coalition Scientific Track runner-up | Team Popeyes

- Led team of 3 to build a playable phase-boundary search game on noisy and error-mitigated PennyLane maps. Code.
- Mapped **ANNNI spin-chain** phases on 8 spins: 92.28% of grid points matched the challenge's reference phase diagram (floating band excluded); **zero-noise extrapolation** cut simulated correlator errors from 0.12-0.18 to 0.002.

Quantum Portfolio Optimization Using QAOA *Apr 2026*

Team Lead, Yale YQuantum Hackathon | Third place

- Led team of 5 on the Markowitz-to-QUBO-to-Ising formulation and QAOA circuit design. Repository.

HONORS AND AWARDS

- **Valedictorian**, B.Tech. graduating class, NMIMS University (2025).
- IBM Qiskit Global Summer School 2026: **Quantum Excellence (Advanced)**. Credential.
- Q-SITE Hacks 2026, in addition to the Scientific Track: **third place**, IBM Quantum Coding Challenge; **third place**, Overall Coding Challenge; **fourth place**, IonQ Championship.
- Crypto Quest Cybersecurity Competition, Tech-Fest Ambiora 2024: **first place**.

INDUSTRY EXPERIENCE

AI and SEO Intern *Feb 2025 - May 2025*

MotionGility, Indore, India

- Built **seven AI-driven content and automation tools** on real-time data and automation workflows, including a video script generator, blog and guest-post generators, a LinkedIn profile finder, and a **sales voice assistant**.

Vulnerability Assessment and Penetration Testing Intern *Oct 2023 - Jan 2024*

Newton's Apple, Pune, India

- Found and helped fix **three critical vulnerabilities** in a confidential Android app; performed static Java code analysis.